

UNIT I

HAZARDS, VULNERABILITY AND DISASTER RISKS

1	What is a disaster? A disaster is a sudden and severe event that causes significant disruption, destruction, or harm to people, property, and the environment. It overwhelms the affected community's ability to cope using its own resources and often requires external assistance to manage the impact and facilitate recovery. Disasters can be natural (e.g., earthquakes, floods) or man-made (e.g., industrial accidents, conflicts).
2	What is a Hazard? A hazard is a potential source of harm or adverse health effect on individuals, property, or the environment. It refers to any condition or situation that has the potential to cause damage, injury, or negative impact. Hazards can be natural (e.g., earthquakes, hurricanes), technological (e.g., chemical spills, industrial accidents), or human-induced (e.g., armed conflicts, terrorism).
3	Define vulnerability in disaster management. In disaster management, vulnerability refers to the susceptibility of individuals, communities, or systems to harm or damage from a disaster. It encompasses the factors that increase the risk of adverse impacts, such as physical, social, economic, and environmental conditions. Vulnerability is influenced by aspects like poverty, inadequate infrastructure, lack of resources, and poor planning, which can limit a community's ability to prepare for, respond to, and recover from disasters.
4	Define Risk and resilience. Risk in disaster management refers to the likelihood of a hazardous event occurring and causing harm, combined with the severity of its impact. It is typically assessed by evaluating the probability of a hazard and the vulnerability of people or systems to that hazard. Resilience is the capacity of individuals, communities, or systems to anticipate, prepare for, respond to, and recover from disasters and adverse events. It involves the ability to withstand disruptions, adapt to changes, and bounce back to a state of normalcy or even improve in the face of challenges.
5	List the different types of disasters. The different types of disasters include: 1. Natural Disasters: Such as earthquakes, floods, hurricanes, tornadoes, volcanic eruptions, and tsunamis. 2. Man-Made Disasters: Including industrial accidents, chemical spills, nuclear accidents, and conflicts or terrorism.
6	Define earthquakes. Earthquakes are sudden and violent shaking of the Earth's surface caused by the release of energy from the movement of tectonic plates along faults or fractures in the Earth's crust. This release of energy generates seismic waves, which can cause ground shaking, surface rupture, and structural damage.
7	Give a few examples of manmade disasters. 1. Industrial Accidents: Such as chemical spills or explosions in factories (e.g., the Bhopal Gas Tragedy in 1984). 2. Nuclear Accidents: Including meltdowns or leaks from nuclear power plants (e.g., Chernobyl disaster in 1986). 3. Oil Spills: Large-scale spills from tankers or drilling rigs (e.g., the Deepwater Horizon oil spill in 2010). 4. Urban Fires: Significant fires in densely populated areas, often due to faulty infrastructure or accidents (e.g., the Grenfell Tower fire in 2017). 5. Terrorist Attacks: Deliberate acts of violence targeting civilians (e.g., the September 11 attacks in 2001).
8	What is a Pandemic?

	A pandemic is an outbreak of a disease that occurs on a global scale, affecting a large number of people across multiple countries or continents. It typically involves a new or highly contagious pathogen that spreads rapidly and widely, leading to significant health, social, and economic impacts. Examples include the COVID-19.
9	List the types of technological disaster. 1. Industrial Accidents: Explosions, fires, or chemical spills in factories or manufacturing facilities (e.g., chemical plant explosions). 2. Transport Accidents: Catastrophic events involving vehicles, trains, ships, or aircraft, such as train derailments or plane crashes (e.g., the Exxon Valdez oil spill). 3. Nuclear Accidents: Incidents involving the release of radioactive materials from nuclear power plants or other facilities (e.g., Chernobyl or Fukushima disasters). 4. Infrastructure Failures: Collapse or malfunction of critical infrastructure like dams, bridges, or power grids (e.g., dam failures or large-scale power outages).
10	Give the socio-economic impact of disaster. The socio-economic impacts of disasters include: 1. Economic Losses: Destruction of property, infrastructure, and businesses leads to significant financial costs and disrupts economic activities. 2. Displacement and Livelihood Disruption: Disasters cause loss of homes and jobs, forcing people to migrate and impacting their ability to earn a living.